

The empirical closed form for OEIS A239847, proved

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Abstract

OEIS A239847 carries an empirical closed form for $a(n)$ — not a recurrence relating consecutive terms but an explicit formula. It is true, and it is decidable rather than empirical. The entry counts arrays subject to a condition on a bounded window of consecutive lines, so the lines of an array are the vertices of a finite digraph, an array is a walk in it, and $a(n)$ is a walk count on $S = 9430$ vertices. Any function of the form $\sum_j p_j(n) \lambda_j^n$ satisfies the monic linear recurrence whose characteristic polynomial is $\prod_j (x - \lambda_j)^{\deg p_j + 1}$; here that polynomial is $q(x) = (x - 1)^{122}$, of degree 122. Two things are then checked exactly: that the walk count satisfies the same recurrence from some index on, which the Cayley–Hamilton theorem reduces to a finite computation, and that the two sequences agree at 122 consecutive indices past that point. A monic recurrence determines every later term from its predecessors, so agreement there is agreement for good.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 The conjecture

OEIS A239847 is “Number of nX5 0..3 arrays with no element equal to one plus the sum of elements to its left or one plus the sum of elements above it or two plus the sum of the elements diagonally to its northwest, modulo 4.”. It has offset 1 and begins

6, 64, 777, 9066, 94613, 874352, 7359682, 57010666, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical polynomial of degree 121 (see link above)

Empirical: $a(n) = (1/7227052254708431910430020397722634669616806306336621491480144943111586253$
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Theorem 3. $a(n) = f(n)$ for every $n \geq 99$.

Proof. The vector $w' = q(M)w$ was formed by 122 matrix–vector products in exact integer arithmetic, and $u^\top M^j w'$ evaluated for $j = 0, 1, \dots$, again exactly. Those integers vanish from the point corresponding to $n = 221$ onwards, and the run was continued until the working vector was identically zero, which settles every larger j at once. So a satisfies (1) for every $n > 220$. By Lemma 1 f satisfies (1) identically. Both were then evaluated in exact arithmetic at the 122 consecutive indices $n = 221, \dots, 342$ and agree at each. A monic recurrence of order 122 determines $a(m)$ from $a(m-1), \dots, a(m-122)$, and for every $m \geq 343$ both $m > 220$ and $m-122 \geq 221$ hold, so induction on m gives $a(m) = f(m)$ for every $m \geq 221$. Below $n = 221$ the recurrence argument gives nothing, since (1) is only established for a from $n = 221$ on. The remaining indices $n = 99, \dots, 220$ were therefore checked one at a time, directly against the walk count in exact integer arithmetic, and agree. $\square$

The range is tight in the sense that was checked: at $n = 98$ the closed form and the sequence differ, so no larger range is available.

5 Verification

Four checks, each independent of the algebra above.

First, the walk model reproduces every one of the 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the model is built by reading the entry’s English, and a misreading gives a different digraph and different counts.

Second, the closed form was evaluated in exact rational arithmetic — not floating point — at every published term from $n = 99$ on, and agrees with the entry’s own DATA at each.

Third, the annihilation test of Lemma 2 was carried out exactly, and the model and the closed form were then compared directly at sixty consecutive indices beyond $n = 99$, far past where the proof already settles the matter.

Fourth, the closed form was deliberately perturbed — by a constant and by a term linear in n — and each perturbed form was rejected by the same comparison, so the test is not one that anything would pass.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A239847.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7; C -finite sequences, Section 4.1.)
- [3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (Closed forms and linear recurrences with constant coefficients.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.